

Symbolic rule synthesis: enumerate, decide, solve

Second Proposal

1. **Enumerate** representative pairs (L, R) within different equivalence groups in concrete synthesis

2. For each candidate, **decide** whether a unitary S exists.

Expensive!

3. If it exists, **solve** the equation $L;S = S;R$

Only need to solve for candidates we know the solution exists

Decide whether a unitary S exists.

FOR UNITARY CIRCUITS L AND R

EIGENVALUE TEST · Lemma 5.3

Necessary and sufficient; expensive

$$\exists \text{ unitary matrix } S : SL = RS \quad \Leftrightarrow \quad \text{eig}(L) = \text{eig}(R)$$

TRACE FILTER · Lemma 5.4

Necessary, not sufficient; more cheap

$$\exists \text{ unitary matrix } S : SL = RS \quad \Rightarrow \quad \text{tr}(L) = \text{tr}(R)$$